

Tiny ancient fossils may reveal an early magnetic navigation system

From: ScienceDaily Strange & Offbeat Mined: 2026-10-03 3 min read

Some animals can find their way using Earth's magnetic field. Now tiny **fossils** may offer clues about how long this ability has existed.

Researchers studied magnetic **fossils** dating back 97 million years. They were found in ancient seafloor **sediments** in the North Atlantic Ocean. The **fossils** have shapes like spearheads, spindles, bullets and needles. Each is no larger than a bacterial cell.

Scientists think a living **organism** made these structures, but they do not know which **organism** it was. Their purpose has also been uncertain.

A team from the University of Cambridge and the Helmholtz Zentrum Berlin made the first three-dimensional images of the magnetic structures inside the **fossils**. The results suggest that the structures were well suited to detecting both the direction and strength of Earth's magnetic field.

That information could have helped an ancient animal work out where it was and which way to travel. **Navigation** means finding your location and your way from one place to another.

Some bacteria already use tiny magnetic **particles** as internal compass needles. But the **fossils** studied here are about 10 to 20 times larger than the **particles** used by bacteria.

Scientists had earlier suggested that these larger structures might have been **protective** spines. The new magnetic images support a different possibility: **navigation**. The creature's identity is still unknown, and the findings may help scientists understand how animals developed the ability to sense Earth's magnetic field.

Word Treasure

fossils -- Preserved remains or traces of life from long ago.

sediments -- Material that settles in layers, such as on the seafloor.

organism -- A living thing.

navigation -- Finding your location and your way between places.

particles -- Very small pieces of material.

protective -- Helping keep something safe from harm.

Background you need

Some animals use Earth's magnetic field to help find their way.

Some bacteria use tiny magnetic particles like internal compass needles.

Break it down

Introduces magnetic sensing and the ancient fossils.

Describes the fossils' location, shapes and unknown maker.

Explains the new images and possible navigation function.

Compares the fossils with bacterial particles and explains the remaining uncertainty.

Why it matters

The fossils may help scientists understand how long animals have used Earth's magnetic field to find their way.

Test what you remember

1. How old are the magnetic fossils researchers studied?

- ☐ (A) About 79 million years old.
- ☐ (B) About 19 million years old.
- ☐ (C) About 37 million years old.
- ☐ (D) About 97 million years old.

2. Where were the fossils found?

- ☐ (A) In ancient North Atlantic seafloor sediments.
- ☐ (B) In ancient Arctic coastal beach sands.
- ☐ (C) In ancient Indian Ocean coral reefs.
- ☐ (D) In ancient South Pacific seafloor sediments.

3. What did the team make images of?

- ☐ (A) The swimming movements of living bacteria.
- ☐ (B) The protective spines of modern animals.
- ☐ (C) The magnetic structures inside the fossils.
- ☐ (D) The sediment layers around ocean islands.

4. What two features of Earth's magnetic field might the structures detect?

- ☐ (A) Its temperature and color.
- ☐ (B) Its speed and brightness.
- ☐ (C) Its direction and strength.
- ☐ (D) Its weight and texture.

5. How does the size of these fossils compare with bacteria's magnetic particles?

- ☐ (A) They are about two to three times larger.
- ☐ (B) They are about 10 to 20 times larger.
- ☐ (C) They are about 10 to 20 times smaller.
- ☐ (D) They are about two to three times smaller.

6. What remains unknown about the fossils?

- ☐ (A) Which organism made them.
- ☐ (B) Whether they have shapes.
- ☐ (C) Which ocean held them.
- ☐ (D) Whether images were made.

Answer key (try the quiz first!): 1: D 2: A 3: C 4: C 5: B 6: A

Questions to think about

MY THOUGHTS (at least 20 words):